Impact of Level 2 EV Charging on Phase Unbalance in Distribution Networks

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***Abstract*—The rapid growth of single-phase EV charging can lead to significant time-varying phase unbalance in distribution grids. In this study, a distribution grid analysis is carried out on a planned Level 2 charger parking lot, located in the Simon Fraser University campus. Historical charging data from existing chargers at the university is used in order to estimate the phase unbalance produced by the new chargers over a 24-hour period. Two types of Level 2 chargers are considered for the new parking lot: a standard level 2 (L2) and a higher voltage level 2 charger with auto transformer (L2T). Both charger scenarios are seen to produce significant unbalanced currents which propagate to the substation and exceed recommended limits regularly throughout the day. This unbalance is seen to be significantly reduced and at times even eradicated by placing a power redistribution converter in the parking lot.**

***Index Terms*—Electric vehicle charging, Level 2 charging, distribution grid analysis, phase unbalance, power redistribution device, power electronics converter.**

I. Introduction

Over the last few years, there has been a noticeable increase in electric vehicles (EV's), representing a total of 14% of new car sales globally in 2022 [1], including both battery EV's (BEV's) and plug-in hybrid EV's (PHEV's). While much of the electric charging demand is being met by home chargers, public electric charger stock has been increasing by 50% each year since 2015 [1].

EV chargers can be categorised as either AC chargers or DC chargers. AC chargers include Level 1 (L1) and Level 2 (L2) chargers which are compatible with both BEV's and PHEV's. The only difference is that L2 chargers provide a higher voltage than L1 chargers, allowing for faster charging. DC fast chargers, also known as Level 3 (L3) chargers, are typically suitable for BEV's only, and require a three-phase power converter. In British Columbia, Canada, 80% of public chargers were L2 (11,500) in 2022, in comparison with L3 chargers [2].

Unlike L3 chargers which are three-phase connected to the grid, L1 and L2 chargers are single-phase connected. High-integration of single-phase chargers, can lead to significant challenges in grid planning and operation [3]. One clear challenge is the power imbalance that may arise, in the three-phase distribution grid when more EV's are charging on one specific phase than the other two phases [4]; this is known as phase unbalance. Traditionally, unbalance caused by known asymmetries could be mitigated by the utility with phase allocation during grid planning. This approach is not sufficient for the phase unbalance caused by EV charging due to the time-varying nature of charging and customer behaviour.

One of the main issues created by phase unbalance is capacity under-utilisation, which occurs when unbalanced feeders become restricted by their most loaded phase. It has become common for utilities to simply reinforce unbalanced networks by investing in new lines and transformers, which comes at a great cost [5].

Phase unbalance also results in the presence of zero- and negative-sequence currents within the grid. Zero-sequence current is proportional to the neutral current which should be equal to zero in a balanced grid. The presence of neutral current is known to cause adverse effects on grid operation, including overheating in the commonly used Δ/Y distribution transformers, as well as causing nuisance tripping by protective relays. Negative-sequence current is also present in an unbalanced grid and is known to poorly affect conventional motors and generators [6].

To mitigate phase unbalance, various methods have been proposed including dynamic phase switching of loads [7], [8], which often requires a large number of switches distributed across a network and typically does not fully eradicate phase unbalance. Power electronics converters can also be used to compensate phase unbalance by redistributing power across the phases [9], these are known as power redistribution devices (PRD's) [10].

In this study, a British Columbia, Canada, distribution grid is analysed for a planned EV charger parking lot located at the Simon Fraser University (SFU) campus. Two possible L2 charger types are studied and historical ChargePoint® data from existing chargers in SFU's campus is used to analyse phase unbalance throughout the day.

The main contributions of this article are summarised as:

1) Phase unbalance analysis is carried out for the EV charger parking lot over a 24-hour period using GridLAB-D and existing ChargePoint® data.
2) A three-phase four-wire (3P-4W) converter is implemented as a power redistribution device in order to mitigate phase unbalance.

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II. EV Charger Modelling

The two types of Level 2 chargers considered for the EV parking lot are the standard L2 charger and a higher voltage charger known as the L2T charger.

A. L2 Charger

The L2 charger is a direct 240 V line-to-line grid connection, as shown in Fig. 1. The EV connects directly to the 240 V provided via the charger. Fig. 1 shows one L2 charger per phase i.e., the line-line connections are Phase A-B, Phase B-C, and Phase C-A. More chargers can be added as represented by the three dots.

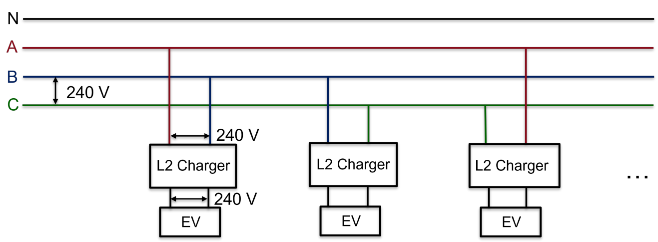

Fig. 1: Electrical diagram for the L2 charger grid connection

B. L2T Charger

A newer Level 2 charger type has been deployed in Canada which connects directly to 600 V distribution lines [11], commonly found in industrial buildings, and multi-unit residential buildings. The advantage of this charger is that the utility transformer required to provide the 240 V line for the L2 chargers is no longer required [11]. However, these new chargers require an internal transformer in order to provide the required 240 V to the EV's, as shown in Fig. 2. These transformers can be solid-state transformers or the auto transformers used in practice in [11]. Fig. 2 shows one L2T charger per phase i.e., the line-neutral connections are Phase A-N, Phase B-N, and Phase C-N.

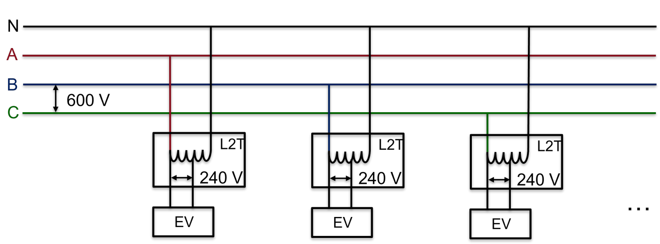

Fig. 2: Electrical diagram for the L2T charger grid connection

III. Grid Analysis

In an EV charging parking lot, it is likely that the charging EV's are not equally distributed across all phases. For example, if more EV's are charging on Phase A than Phase B or C, or if the EV's on one phase are charging at a higher power. This leads to phase unbalance reflected in the currents and voltages of the three-phase four-wire (3P-4W) distribution grids.

A. Sequence Decomposition

Three-phase voltages and currents in the distribution grid can be decomposed into sequence components,

$$\begin{bmatrix} \mathbf{I}_0 \\ \mathbf{I}_+ \\ \mathbf{I}_- \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} \mathbf{I}_a \\ \mathbf{I}_b \\ \mathbf{I}_c \end{bmatrix} \tag{1}$$

where $a = e^{j\frac{2\pi}{3}}$ and $\mathbf{I}$ is current. This current $\mathbf{I}$ can be replaced by $\mathbf{V}$ for the voltage decomposition. In a balanced grid, only positive-sequence components exist. However, in an unbalanced grid, zero- and negative-sequence components can also exist. The neutral line current is directly related to the zero-sequence current as,

$$\mathbf{I}_n = \mathbf{I}_a + \mathbf{I}_b + \mathbf{I}_c = 3\mathbf{I}_0 \tag{2}$$

B. Unbalance Factors

The current and voltage unbalance factors, as defined in IEC 61000-2-2 [12], is the ratio of negative- and zero-sequence component over the positive-sequence component. The current unbalance factor CUF is defined as [13],

$$CUF_- = \frac{|\mathbf{I}_-|}{|\mathbf{I}_+|} \cdot 100\% \tag{3}$$

$$CUF_0 = \frac{|\mathbf{I}_0|}{|\mathbf{I}_+|} \cdot 100\% \tag{4}$$

$$CUF = \sqrt{CUF_-^2 + CUF_0^2} \tag{5}$$

The same principle applies to the voltage unbalance where the overall voltage unbalance factor is,

$$VUF = \sqrt{VUF_-^2 + VUF_0^2} \tag{6}$$

Another index for voltage unbalance is defined in ANSI C84.1 as [14],

$$VU = \frac{\text{Maximum Voltage Deviation}}{\text{Average Voltage}} \cdot 100\% \tag{7}$$

This index only considers the magnitude and is therefore less insightful than the VUF index. ANSI C84.1 states that VU should not exceed 3% [14], and IEC 61000-2-2 states that VUF should not exceed 2% other than in rural areas where 3% may occur [12]. The current unbalance factor limit is not stated in these standards, however BC Hydro, the utility in British Columbia, states that a current unbalance (CU) of 10-20% is considered normal [6]. It is implied in [6] that CU is calculated as,

$$CU = \frac{\text{Maximum Current Deviation}}{\text{Average Current}} \cdot 100\% \tag{8}$$

C. Distribution Grid Modelling

GridLAB-D, an open-source distribution grid modelling tool developed by the U.S. Dept. of Energy, is used for the analysis conducted in this study [15]. GridLAB-D performs three-phase power flow using detailed component models.

IV. Power Redistribution Device

Three-phase four-wire (3P-4W) converters can be used to redistributed real and reactive power across the three phases [13], [16], [17]; this is known as power redistribution. By placing a power redistribution device (PRD) near an unbalanced load, as seen in Fig. 3, the phase unbalance on the grid side can be mitigated and even eradicated depending on the sizing of the converter [10].

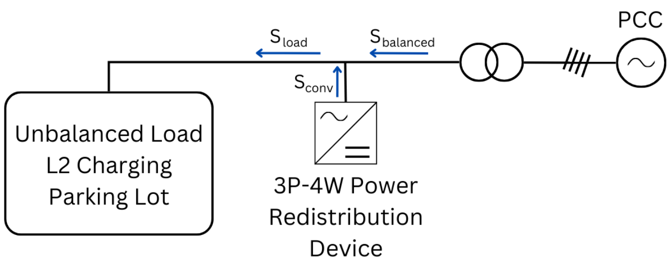

Fig. 3: One-line diagram of 3P-4W network with a PRD shunt connected next to the unbalance load caused by L2 charging.

When the charging load is unbalanced, the 3P-4W converter can be used to balance the load such that,

$$\mathbf{S_{balanced,Ph}} = \frac{1}{3}\Big(\mathbf{S_{load,A}} + \mathbf{S_{load,B}} + \mathbf{S_{load,C}}\Big) \tag{9}$$

where Ph represents each phase A, B and C. Therefore, the required converter power in each phase is,

$$\mathbf{S_{conv,Ph}} = \mathbf{S_{load,Ph}} - \mathbf{S_{balanced,Ph}} \tag{10}$$

Note that ideally, if the converter power loss is negligible then the net power from the converter is zero,

$$\mathbf{S_{conv,A}} + \mathbf{S_{conv,B}} + \mathbf{S_{conv,C}} = 0 \tag{11}$$

hence, a PRD does not require a DC power supply to operate [17]. If the converter size is inadequate the phase unbalance can be reduced but not entirely eradicated.

V. Case Study

An EV charger parking lot is planned at the SFU campus, comprising of 20 dual-port 22 kW Level 2 chargers; hence, a total of 40 charging spaces. The chargers are expected to be distributed across each phase, as shown in Fig. 4.

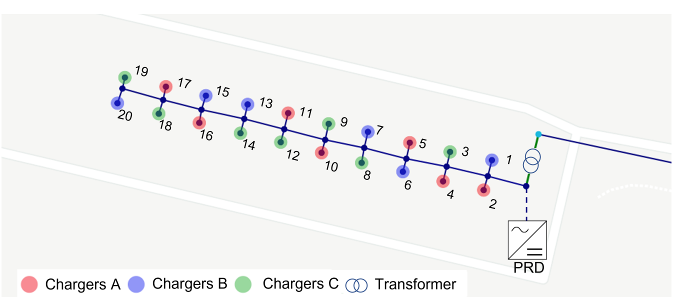

Fig. 4: Grid connections showing the EV chargers connected to specific phases and the PRD connection at the transformer.

A 500 kVA Wye/Wye utility transformer is required to step-down the 12.7 kV distribution voltage to the required input voltage of the chargers. The electrical properties for the L2 and L2T chargers considered are outlined in Table I. Since, the L2 charger is a direct electrical connection to the EV, the power factor is equal to the EV's on-board charger power factor which is typically close to unity. However, the power factor for the L2T is taken to be 0.8 which is caused by the auto transformer. In the "L2 scenario" all chargers will be L2 chargers, and in the "L2T scenario" all chargers will be L2T. Other loads present in the grid are removed and only the impact of the chargers is analysed.

TABLE I: L2 and L2T chargers' electrical properties

	L2 Charger	L2T Charger
Input voltage (V)	240	600
Output voltage to EV (V)	240	240
Power Factor	0.99	0.8

The PRD used to compensate phase unbalanced is placed on the secondary side of the transformer as shown in Fig. 4. The PRD here is rated at 50 kVA which is chosen to be of a similar size as a DC-Fast charger.

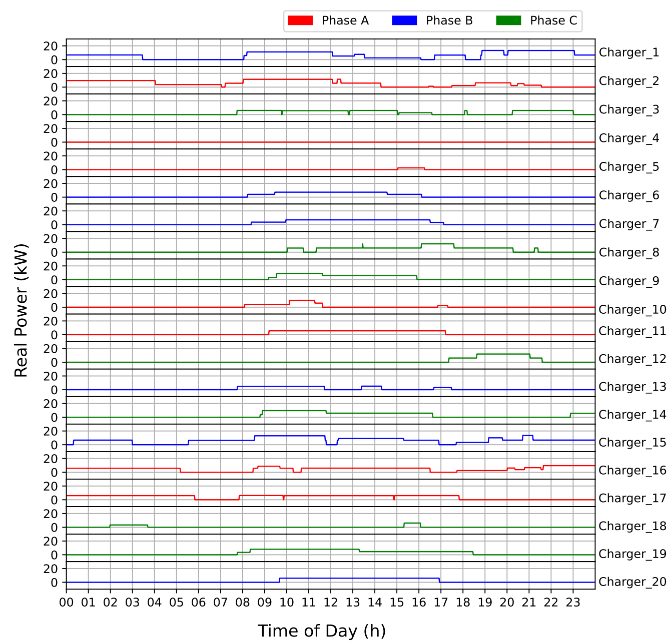

Fig. 5: Daily charging profiles from SFU Chargepoint chargers imposed on Chargers 1-20 and divided into Phases.

A. Typical Daytime Charging Profiles

To analyse the phase unbalance caused by the EV charger parking lot, each charger, shown in Fig. 4, is appointed a 24-hour charging profile shown in Fig. 5. These charging profiles are taken from historical ChargePoint® data from existing 14.4 kW L2 chargers distributed around the SFU campus and have been scaled to the rating of the new chargers (22 kW).

VI. Results

Based on the charging profiles, Fig. 5, the current and voltage unbalance is calculated by running hourly grid analysis using GridLAB-D for the L2 and L2T scenarios.

A. Daytime scenario for L2 Chargers

In the L2 scenario, phase unbalance produces negative-sequence current, as shown in Fig. 6, however zero-sequence current remains zero, since the neutral line is not used in the L2 charger connection, Fig. 1. Although the high charging load starts from 9 AM, when consumers arrive at work, the negative-sequence current is not following the same pattern, showing that the phase unbalance is not necessarily related to loading. Fig. 7 (a) shows that the highest CUF occurs between 4 AM - 5 AM, although the current magnitudes at this time are low, as seen in Fig. 6. This maximum occurs because the only chargers used during this time are Chargers 16 and 17 which are both Phase A, and the other phases are unused, as seen in Fig. 5. The alternative current unbalance index CU is seen to exceed the 10-20% unbalance norm stated by BC Hydro [6], for the majority of the day.

Note that in Fig. 7 (a), CUF is equal to CUF_{-} since there is no zero-sequence current. The voltage unbalance factors are found to never exceeds the 2% limit.

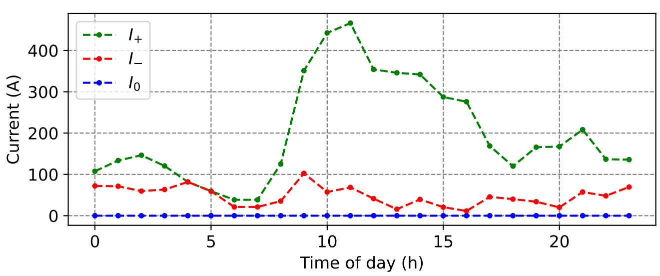

Fig. 6: Sequence-currents through the utility transformer over 24-hours due to EV charging via the L2 chargers.

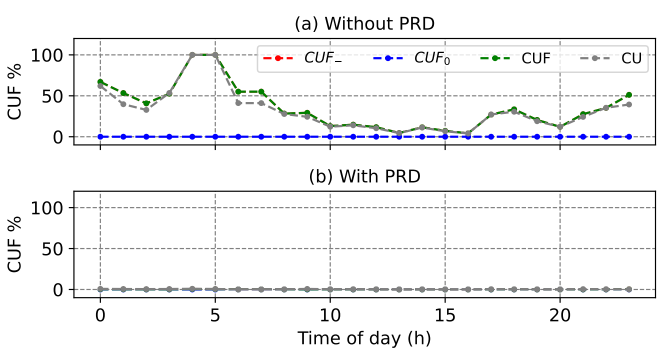

Fig. 7: Current unbalance factors at the utility transformer over a 24-hour period due to EV charging via the L2 chargers, without and with the 50 kVA PRD in (a) and (b) respectively.

When using the 50 kVA PRD it is found that phase unbalance is successfully eradicated throughout the day as reflected in the current unbalance factors seen in Fig. 7 (b). Complete power redistribution cannot be achieved in such a way using dynamic phase switching presented in [7], [8], due to the discrete nature of switching entire loads from one phase to another.

B. Daytime scenario for L2T Chargers

For the L2T scenario, zero-sequence current also exists and is equal to negative-sequence current, as shown in Fig. 8. This is caused by having the L2T chargers connected phase-to-neutral, as seen in Fig. 2. The highest unbalanced currents occur again at 9 AM, Fig. 8, which is found to result in a substation neutral current of 8.0 A.

When the 50 kVA PRD is added, as seen in Figs. 8-9, phase unbalance is seen to be significantly reduced and even eradicated for most of the day. However, 50 kVA is insufficient to eradicate phase unbalance entirely. This is due to the significant reactive power drawn by the L2T auto transformers. From the real and reactive power, shown in Fig. 10, it can be seen that real power is mostly balanced in Fig. 10 (b) since the PRD prioritised real power balancing. However, reactive power unbalance still exists, indicating that a PRD with higher rating is required for full phase unbalance compensation in this case.

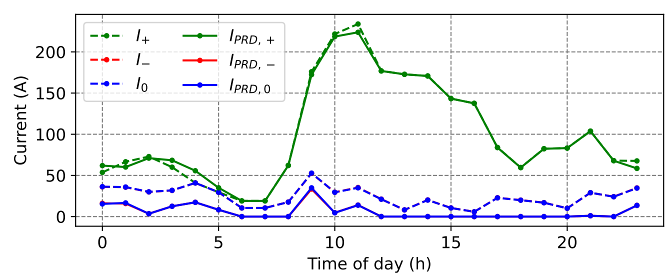

Fig. 8: Unbalanced Sequence-currents through the utility transformer over a 24-hour period due to EV charging via the L2T chargers, without and with the 50 kVA PRD.

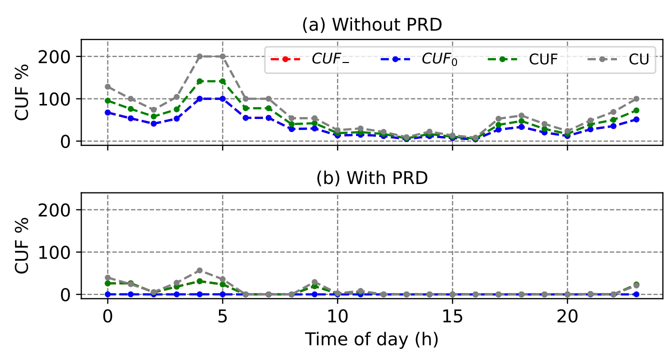

Fig. 9: Current unbalance factors at the utility transformer over 24-hours due to EV charging via L2T chargers, without and with the PRD in (a) and (b) respectively.

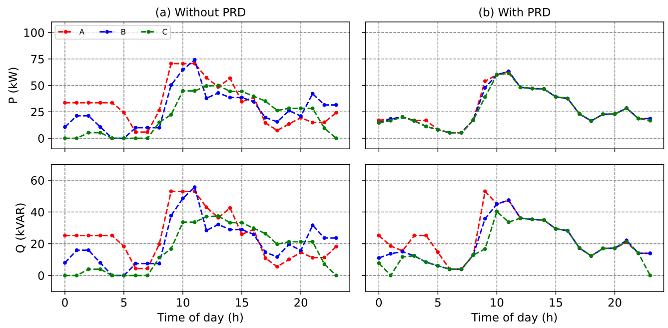

Fig. 10: Real and reactive power through the utility transformer in the L2T scenario, without and with the PRD in (a) and (b) respectively.

VII. Discussion

The worst unbalance case scenario requires 14 EV's to be charging at full power on Phase A only, which has a small probability of happening. However, it would lead to a substation negative-sequence current of 9.5 A for both charger types and a substation neutral current of 28.8 A (for the L2T). Even in the typical charging scenario which uses existing charger data, Sections VI.A-VI.B, it is seen that phase unbalance exceeds allowable current limits, and varies significantly throughout the day. Similar findings were found in a European grid study [4], which also includes the effect of single-phase rooftop solar PV.

From this study, it is clear that the line-to-line connection of chargers, rather than the line-to-neutral is greatly beneficial to reduce unbalance, since only negative-sequence current arises. Although the L2T charger has the added benefit of not requiring a utility transformer when a 600 V network exists, the auto transformer used leads to significant reactive power loading, and the line-to-neutral connection, also seen in Level 1 home chargers, can easily lead to neutral currents propagating through the network.

It is clear that active grid balancing solutions are required when high levels of single-phase EV charging is integrated into the grid. When a PRD is used in the parking lot, phase unbalance on the grid side is seen to be significantly reduced and often is fully eradicated. A promising solution would be to allow DC-fast chargers to behave as PRD's when they are not being actively used for EV charging. DC-fast chargers placed next to Level 2 chargers could provide phase balancing support, and more generally could be placed in locations in the grid prone to phase unbalance. DC-fast chargers typically use standard three-phase three-wire (3P-3W) converters which are sufficient to compensate the negative-sequence current that results from line-to-line charging (e.g. L2) [17], since neutral current does not exist in this case.

VIII. Conclusion

Here, a Level 2 charger parking lot analysed using 24-hour historical data, indicates that single-phase charging results in significant levels of phase unbalance that vary throughout the day. The current unbalance limits are exceeded for the majority of the 24-hour period and single-phase charging is seen to produce negative-sequence currents, and even neutral currents for chargers connected line-to-neutral (e.g. the L2T). These harmful currents propagate to the substation demonstrating that active solutions are required. The power redistribution device placed alongside the EV charger parking lot is seen to significantly reduce and even eradicate phase unbalance.

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